SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Questions

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

QUESTIONS: 5 **Max Marks50** **Duration :60 Minutes**

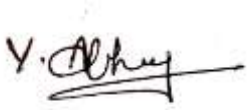
Annexure A

Scenario-Based Questions

Code of Conduct:

I Y. Abhay- 192411219 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Scenario-Based Questions

1. **Dental X-Ray Enhancement and Cavity Region Extraction**

A dental clinic uses a digital radiography system to help dentists locate early-stage cavities (caries) in teeth. The X-ray images are dark and low in contrast, with quantum noise and overlapping tooth structures, so small cavity regions blend into the surrounding enamel and dentine.

Question: Apply suitable grey-level transformations (such as negative, log or power-law transformation) and histogram equalization to improve the visibility of the cavity regions. Select an appropriate spatial smoothing filter to suppress noise without blurring the cavity boundaries. Subsequently, apply thresholding or region growing to extract the suspected cavity region, and use edge detection to outline it. Explain the sequence of processing steps and justify the choice of each technique.

2. **Night-Time Vehicle Number Plate Region Extraction**

A traffic police department installs CCTV cameras at a toll gate to capture vehicles at night. The images suffer from very low brightness, headlight glare, sensor noise, and motion blur. The system must locate and extract the number-plate region so that the characters can be read in a later stage.

Question: Apply suitable grey-level transformations and histogram processing to enhance the dark regions while controlling glare. Use appropriate smoothing and sharpening filters (for example, median filtering followed by Laplacian or unsharp masking) to reduce noise and restore character edges. Apply edge detection, line detection and edge linking to locate the rectangular plate boundary, and use thresholding to separate the characters from the plate background. Present the complete processing pipeline and explain how each step contributes to extracting the number-plate region.

3. **Textile Fabric Defect Detection Using Frequency-Domain Processing**

A textile mill inspects woven fabric rolls using a line-scan camera. The fabric has a regular, repeating weave pattern, and defects such as holes, broken threads, oil stains and knots disturb this pattern. The captured images also contain periodic noise from the machine lighting and uneven illumination across the width of the roll.

Question: Apply suitable frequency-domain filtering techniques (low-pass, high-pass, or notch/band-reject filtering) to suppress the periodic noise and the regular weave texture, so that the defects stand out. Use histogram processing to correct the uneven illumination. Subsequently, apply thresholding and boundary detection to segment and outline the defective regions. Explain the order of operations and justify why frequency-domain processing is preferred over spatial-domain processing in this scenario.

4. **Blood Cell Segmentation in Microscopic Images**

A pathology laboratory uses an automated microscope to capture images of stained blood smears for counting red and white blood cells. The images show uneven staining, poor contrast between cells and background, touching or overlapping cells, and noise from dust particles on the slide.

Question: Apply suitable contrast enhancement and histogram processing techniques to improve the separation between cells and background. Use an appropriate spatial filter to remove small noise particles while preserving cell boundaries. Apply global or adaptive thresholding to separate cells from the background, and use region-based segmentation (region growing or region splitting and merging) to separate touching cells. Finally, apply edge detection and boundary detection to outline individual cells. Explain the processing sequence and the role of each technique in obtaining accurate cell regions.

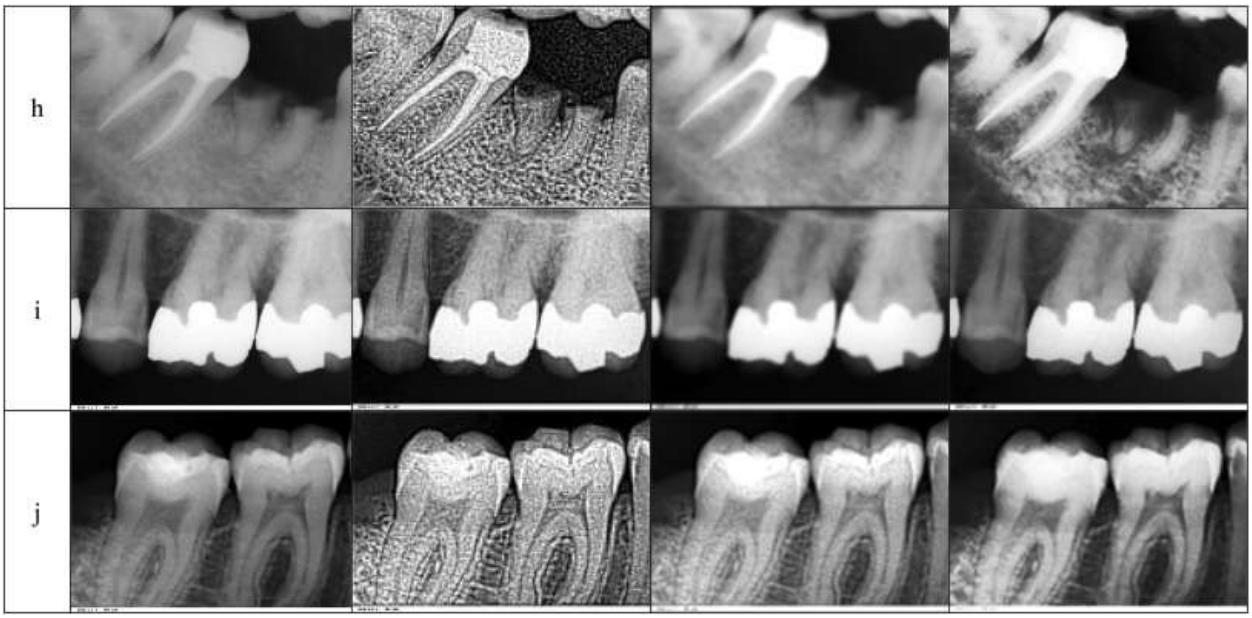
5. **Text Extraction from Degraded Historical Documents**

A state archive is digitising old handwritten manuscripts. The scanned pages contain yellowed paper, ink bleed-through from the reverse side, stains, faded strokes and uneven lighting across the page. The archive requires clean extraction of text regions before handwriting recognition.

Question: Apply suitable grey-level transformations and histogram processing to improve the contrast between faded ink and the background. Select appropriate smoothing and sharpening filters to remove stains and noise while strengthening the pen strokes. Compare global thresholding (such as Otsu's method) with local (adaptive) thresholding for this scenario and apply the more suitable one to separate text from the background. Use edge detection and boundary detection to identify text lines and word regions. Explain the sequence of operations and justify each decision.

Answers:

1. Dental X-Ray Enhancement and Cavity Region Extraction



Introduction

Dental X-ray images are often dark, low in contrast, and affected by quantum noise. Early-stage cavities may blend with the surrounding enamel and dentine. Image enhancement and segmentation techniques can improve cavity visibility and help dentists identify suspected regions.

Processing steps

Step 1: Input dental X-ray image

Step 2: Grey-level transformation

Gamma correction or logarithmic transformation

Step 3: Histogram equalization

Improve contrast between tooth structures and cavities

Step 4: Noise removal

Median or edge-preserving bilateral filtering

Step 5: Segmentation

Adaptive thresholding or region growing

Step 6: Edge detection and cavity outlining

Canny edge detector

Output: Enhanced X-ray with suspected cavity regions outlined

Explanation and justification

1. Grey-level transformation: Apply gamma correction with \(\gamma<1\) to brighten dark regions. Logarithmic transformation can also expand the visibility of lower-intensity details. A negative transformation may be used if reversing the image intensities makes the cavity easier to distinguish.

2. Histogram equalization: Redistribute intensity values to improve overall contrast. CLAHE may be preferred when local contrast improvement is needed without excessive enhancement of noise.

3. Noise removal: Apply a median filter to suppress isolated noise while preserving sharp boundaries. A bilateral filter is another option because it smooths similar-intensity areas while retaining edges.

4. Segmentation: Apply adaptive thresholding when intensity varies across the X-ray. Alternatively, use region growing by selecting a seed point in a suspected cavity and adding neighboring pixels with similar intensity.

5. Edge detection: Use the Canny operator to detect intensity changes and outline the segmented cavity region.

6. Post-processing: Remove small unwanted regions and retain plausible cavity-shaped components. The result should be reviewed by a dentist.

Conclusion

Combining grey-level transformations, histogram equalization, edge-preserving smoothing, segmentation, and edge detection improves cavity visibility and produces outlined candidate regions. These regions support clinical assessment but do not independently establish a diagnosis.

2. Night-Time Vehicle Number Plate Region Extraction



Introduction

Night-time CCTV images often have low brightness, headlight glare, sensor noise, and motion blur. These problems make number-plate boundaries and characters difficult to identify. Image enhancement, filtering, edge detection, and segmentation can be combined to locate and extract the plate.

Processing steps

1. Input night-time vehicle image

2. Gamma correction and CLAHE

3. Median filtering and sharpening

4. Canny edge detection

5. Hough line detection and edge linking

6. Rectangular plate identification and cropping

7. Adaptive thresholding for character separation

Output: Extracted number plate ready for character recognition

Explanation and justification

1. Grey-level transformation: Apply gamma correction with \(\gamma<1\) to brighten dark regions. This improves the visibility of the vehicle and its number plate. However, excessive brightening may amplify headlight glare.

2. Histogram processing: Use CLAHE (Contrast Limited Adaptive Histogram Equalization) to enhance local contrast. Its contrast limit helps control excessive enhancement around bright headlights.

3. Noise removal: Apply a median filter to remove isolated noise while preserving edges. This improves the reliability of later edge detection.

4. Sharpening: Apply unsharp masking or a Laplacian-based sharpening filter to strengthen plate boundaries and character edges. Sharpening should be moderate because it can amplify noise. Severe motion blur may require a separate deblurring method.

5. Edge detection: Apply the Canny edge detector to identify strong intensity changes around the plate.

6. Line detection and edge linking: Use the Hough transform to identify horizontal and vertical lines. Connect nearby broken edges using morphological closing or edge-linking techniques to form a rectangular boundary.

7. Plate extraction: Identify candidate rectangles using aspect ratio, size, and edge density. Select a suitable candidate and crop the plate region.

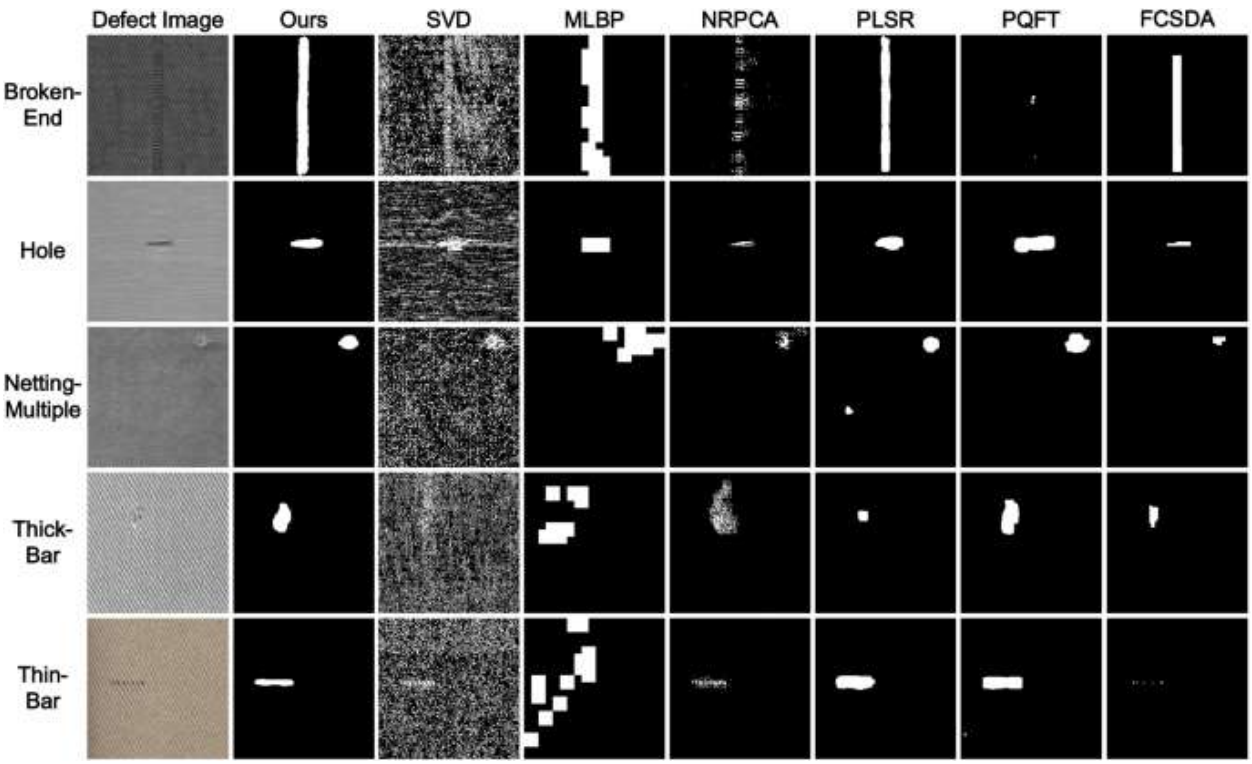
8. Thresholding: Apply adaptive thresholding to separate characters from the plate background. If the plate has uniform illumination, Otsu's thresholding can also be used.

9. Final refinement: Remove small unwanted components and correct the plate's orientation before passing it to an optical character recognition (OCR) system.

Conclusion

The combination of gamma correction, CLAHE, noise filtering, sharpening, edge detection, and line detection helps locate number plates in night-time images. Thresholding then separates the characters from the plate background, making the cropped region suitable for OCR.

3. Textile Fabric Defect Detection Using Frequency-Domain Processing



Introduction

Woven fabric contains a regular repeating texture, while defects such as holes, broken threads, oil stains, and knots disturb this pattern. Periodic lighting noise and uneven illumination further complicate inspection. Frequency-domain filtering is useful because repeating textures and periodic noise often appear as identifiable frequency components.

Processing steps

1. Input fabric image

2. Correct uneven illumination

3. Apply Fourier Transform (FFT)

4. Apply notch/band-reject and texture-suppression filters

5. Apply inverse Fourier Transform

6. Contrast enhancement and thresholding

7. Boundary detection and defect identification

Output: Fabric image with defective regions segmented and outlined

Explanation and justification

1. Illumination correction: Estimate the slowly varying background and normalize or subtract it to reduce brightness variations across the fabric width.

2. Fourier Transform: Convert the image from the spatial domain into the frequency domain using the Fast Fourier Transform (FFT). This reveals the frequency components associated with periodic patterns and noise.

3. Notch filtering: Apply notch-reject filters to remove specific frequency peaks caused by periodic lighting noise.

4. Texture suppression: Use carefully designed band-reject filters to attenuate dominant weave frequencies. High-pass filtering may emphasize fine defects, while low-pass filtering may suppress high-frequency noise. Neither should be applied indiscriminately because actual defects may occupy the same frequency ranges.

5. Inverse Fourier Transform: Convert the filtered frequency representation back into a spatial image for further processing.

6. Histogram processing: Apply contrast stretching or CLAHE to make defects more distinguishable from the surrounding fabric.

7. Thresholding: Separate pixels representing possible defects from the regular fabric background. Adaptive thresholding is useful when residual illumination varies locally.

8. Boundary detection: Use morphological operations, connected-component analysis, and edge detection to outline suspected defective regions and remove tiny unwanted components.

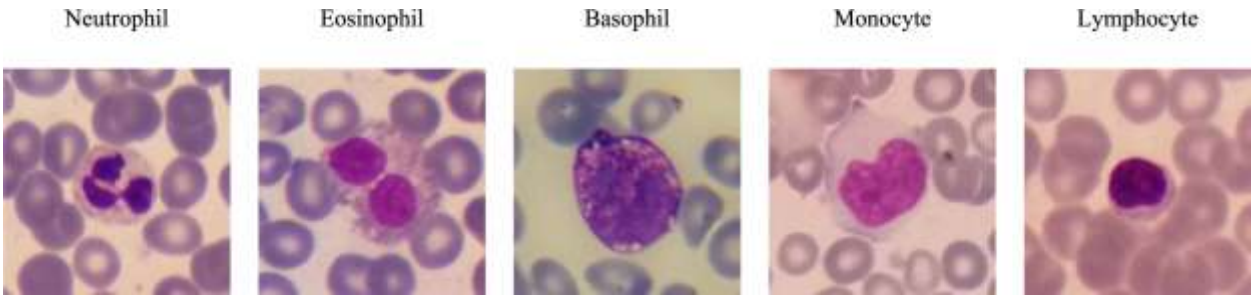
Why is frequency-domain processing preferred?

- Regular weave patterns and periodic noise create recognizable peaks in the frequency spectrum.
- Notch filters can selectively remove unwanted periodic components.
- Spatial smoothing may blur both the weave and the small defects.
- Frequency-domain filtering provides direct control over selected frequency ranges.

Conclusion

Frequency-domain processing is effective for suppressing periodic lighting noise and dominant weave patterns. After illumination correction, frequency filtering, contrast enhancement, and segmentation, defective regions can be outlined for automated textile inspection. Careful filter design is essential to avoid removing defect details along with the fabric texture.

4. Blood Cell Segmentation in Microscopic Images



Introduction

Microscopic blood-smear images are used to identify and count red blood cells (RBCs) and white blood cells (WBCs). Uneven staining, low contrast, dust particles, and touching cells make segmentation difficult. Image enhancement, filtering, thresholding, and region-based segmentation can improve the extraction of individual cells.

Processing steps

1. Input microscopic blood-smear image

2. Contrast enhancement using CLAHE

3. Noise removal using median filtering

4. Global or adaptive thresholding

5. Region-based segmentation and watershed

6. Edge detection and boundary extraction

Output: Individual cell regions outlined for counting

Explanation and justification

1. Contrast enhancement: Apply CLAHE to improve local contrast between cells and the background. It enhances local details while limiting excessive contrast amplification.

2. Histogram processing: Use histogram stretching or equalization to improve the distribution of intensity values. For stained images, processing an appropriate grayscale channel or color component may improve cell-background separation.

3. Noise removal: Apply a median filter to suppress isolated dust particles and small noise spots. A small filter window helps preserve the cell boundaries and fine details.

4. Thresholding: Apply adaptive thresholding when staining and illumination vary across the image. Global thresholding, such as Otsu's method, is suitable when the cell and background intensities are sufficiently distinct and relatively uniform.

5. Region-based segmentation: Use region growing to group neighboring pixels with similar intensity or color, starting from selected seed points. Alternatively, use watershed segmentation to separate touching cells.

6. Separating touching cells: Calculate a distance transform from the binary cell mask and identify cell centers as markers. Apply marker-controlled watershed to divide connected cell regions into individual cells.

7. Edge detection: Apply the Canny operator or gradient-based edge detection to identify cell boundaries.

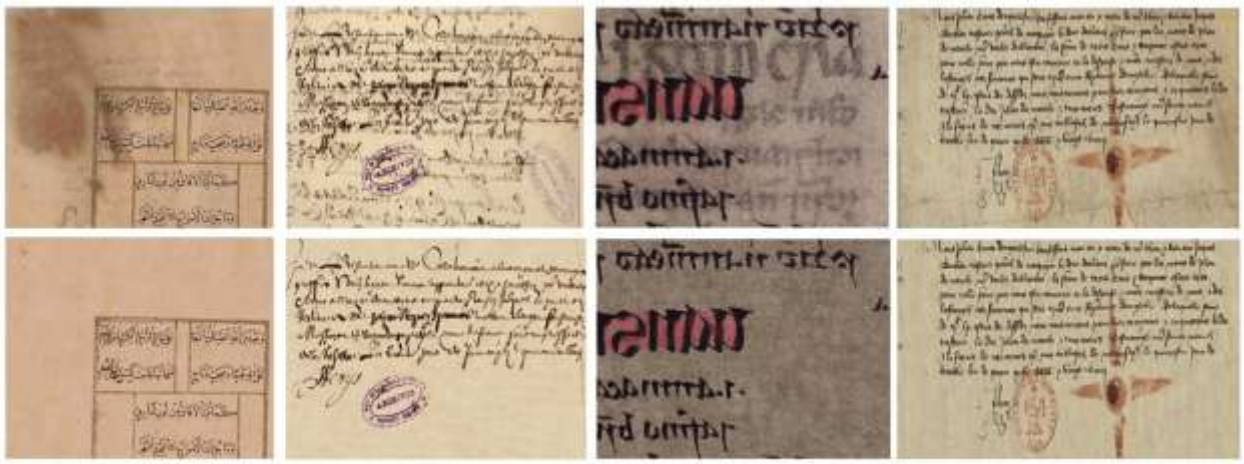
8. Boundary extraction: Use connected-component analysis and morphological operations to outline each cell, remove small unwanted regions, and calculate properties such as area and perimeter.

9. Cell counting: Count the segmented components and classify cells using suitable features such as size, shape, and staining characteristics.

Conclusion

CLAHE improves cell visibility, median filtering removes noise, and thresholding separates cells from the background. Region-based segmentation, particularly marker-controlled watershed, helps separate touching cells. Edge and boundary detection then support individual cell counting and analysis.

5. Text Extraction from Degraded Historical Documents



Introduction

Historical manuscripts often contain yellowed paper, faded handwriting, ink bleed-through, stains, and uneven illumination. These conditions make it difficult to distinguish ink strokes from

the background. Image enhancement, background correction, filtering, and adaptive thresholding can improve text extraction before handwriting recognition.

Processing steps

1. Input scanned manuscript

2. Grayscale conversion and grey-level transformation

3. CLAHE and background correction

4. Noise removal and stroke enhancement

5. Adaptive thresholding

6. Edge detection and connected-component analysis

Output: Extracted text lines and word regions for handwriting recognition

Explanation and justification

1. Grayscale conversion: Convert the scanned page to grayscale to simplify intensity-based processing. If the yellow paper or colored stains contain useful information, color-based background correction can be performed before grayscale conversion.

2. Grey-level transformation: Apply gamma correction or a logarithmic transformation to improve the visibility of faded ink strokes. The transformation should enhance the ink without making the background too prominent.

3. Histogram processing: Apply CLAHE to enhance local contrast between the handwriting and paper. Unlike basic global histogram equalization, CLAHE adapts to local regions and limits excessive enhancement.

4. Background correction: Estimate the slowly varying paper background and subtract or normalize it. This reduces uneven illumination and some broad stain effects. Bleed-through may require additional background estimation or specialized suppression techniques.

5. Noise removal: Apply a small median filter to remove isolated spots. Avoid strong smoothing because it may erase thin or faded handwriting strokes.

6. Sharpening: Use unsharp masking to strengthen faint pen strokes. Apply it carefully because excessive sharpening can amplify stains, bleed-through, and scanning noise.

7. Thresholding: Convert the enhanced image into foreground text and background pixels.

Global versus local thresholding

Global thresholding (Otsu's method)	**Local thresholding (Adaptive/Sauvola)**
Calculates one threshold for the entire image.	Calculates thresholds for small local regions.
Works well when illumination is relatively uniform.	Works well with uneven lighting and stained paper.
May lose faded strokes or retain background stains.	Better adapts to local changes in paper brightness.
Simpler and computationally efficient.	Requires selecting an appropriate neighborhood size and parameters.

Selected technique: Adaptive thresholding

Adaptive thresholding, particularly Sauvola thresholding, is more suitable because historical manuscripts contain faded strokes and uneven backgrounds. It determines the threshold using local intensity statistics, improving the separation of ink from paper.

8. Edge detection: Apply Canny edge detection when useful to identify the boundaries of connected ink strokes. However, edge detection alone cannot reliably separate all handwriting from stains.

9. Boundary detection: Use connected-component analysis to identify text components. Horizontal projection profiles can locate text lines, while component positions and spacing can help group characters into words.

10. Final refinement: Remove obvious isolated noise, preserve thin strokes, and prepare the extracted text regions for handwriting recognition.

Conclusion

Grey-level transformations and CLAHE improve the visibility of faded ink. Background correction reduces uneven illumination, while careful smoothing and sharpening remove noise and strengthen strokes. Adaptive thresholding is preferred over global Otsu thresholding because the document contains local variations in brightness. Finally, edge detection and connected-component analysis help extract text lines and word regions for handwriting recognition.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand